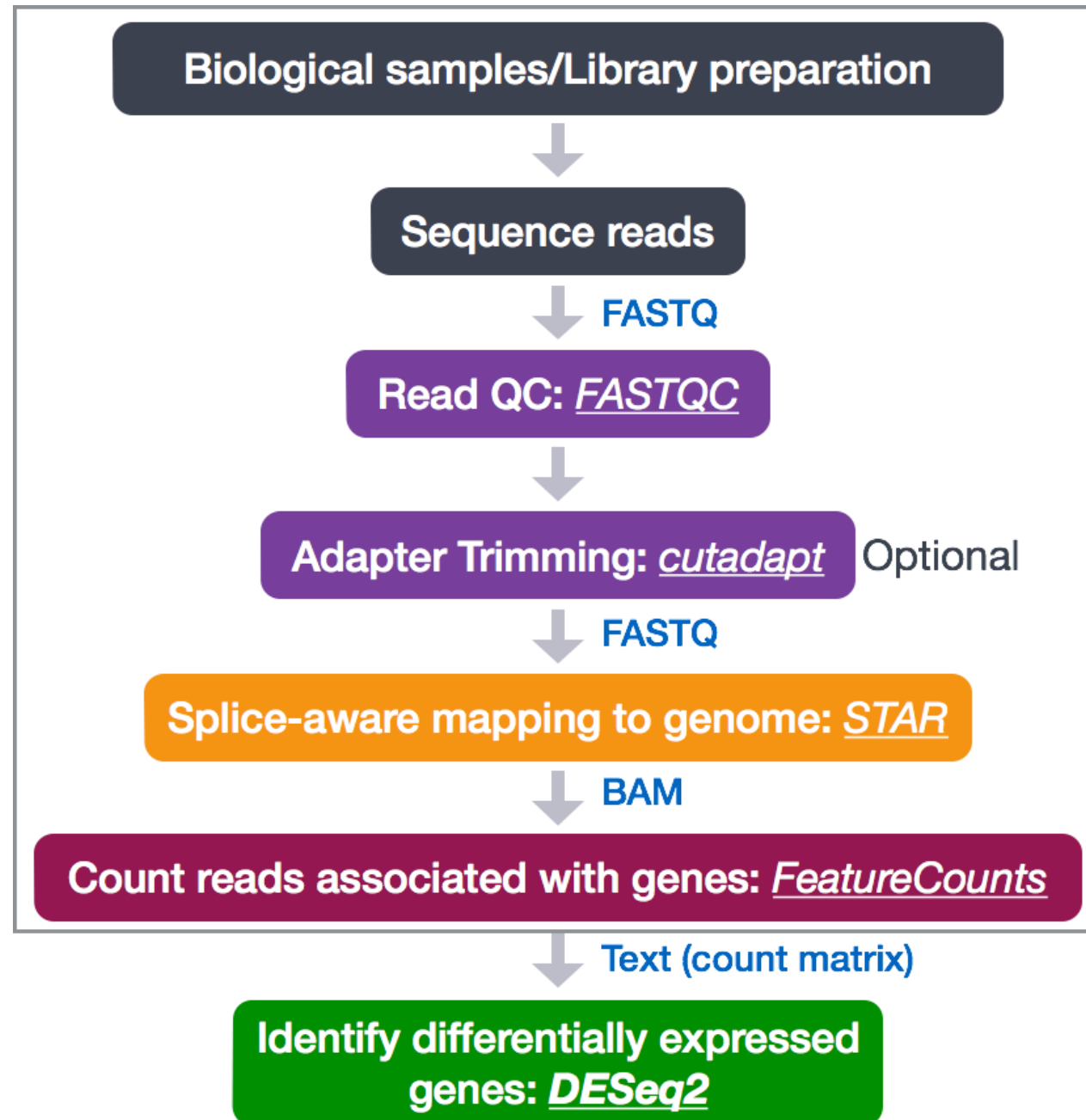
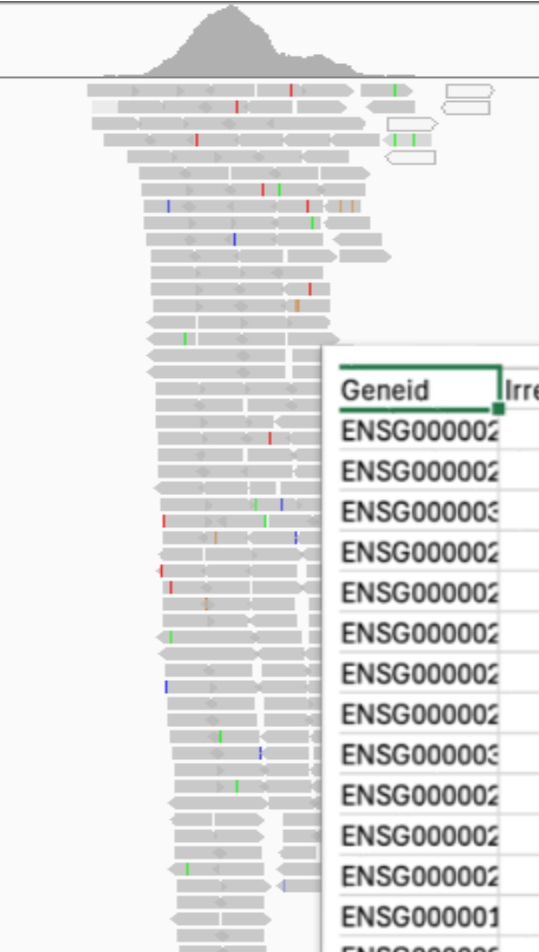




@SRR948304.5141 UNC14-SN744:253:D135LACXX:5:1101:4054:4287
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+
CCCCFFFFHHHHGHIIJJJ:
@SRR948304.5160 UNC14-
CCACGAAGTCAAGATCGCCGA
+
BCCCCFFHHGHJJJJJJJ:
@SRR948304.5196 UNC14-
GGAACGCCTACTGGTACTATC
+
CCCCFFFFHHHJGHIIJJJ:
@SRR948304.5200 UNC14-
CAGGAAGAAGGAGTAGTCCAT
+
@@CFFBDDHHBHIIHIFGHII
@SRR948304.5215 UNC14-
CCTTCTCAACGACTACTACA
+
@@FFFFFHGDHFIHGIJJFI
@SRR948304.5247 UNC14-
CTACAGCTTCCGCAAGAACTA
+
BB@FFFFHHHFIJJJJJJJ:
@SRR948304.5250 UNC14-
CCTGGTTGAACGCTAGATAT
+
CCCCFFFFHHHJGHIIJJJ:



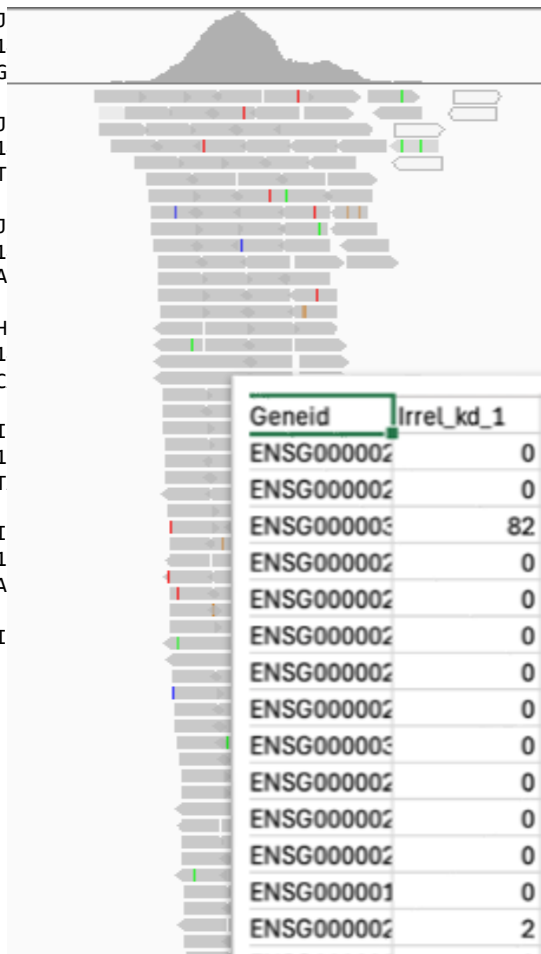
A few weeks ago, we were here, with a matrix of counts

Geneid	lrrel_kd_1	lrrel_kd_2	lrrel_kd_3	Mov10_oe_1	Mov10_oe_2	Mov10_oe_3
ENSG000002	0	0	0	0	0	1
ENSG000002	0	0	0	0	0	0
ENSG000003	82	86	65	119	106	52
ENSG000002	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000002	0	2	1	0	3	3
ENSG000002	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000003	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000001	0	0	0	0	0	0
ENSG000002	2	0	0	2	1	0
ENSG000002	0	0	0	0	0	0
ENSG000003	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
FNSG000000	1	1	0	0	1	0

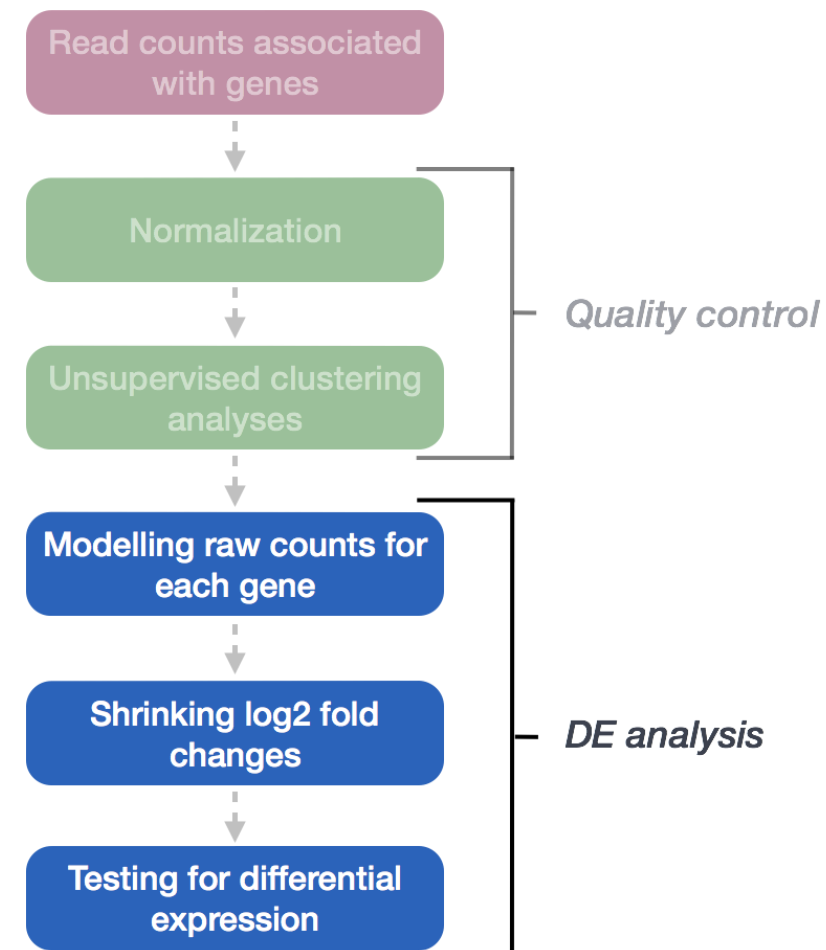
```

@SRR948304.5141 UNC14-SN744:253:D135LACXX:5:1101:4054:4287
AGGACTTTGGCTGTCCCAACCGTACAAGTGGGTCTCTCCCTCATGGA
+
CCCCFFFFHHHHGHIIJJ
@SRR948304.5160 UNC1
CCACGAAGTCAAGATCGCCG
+
BCCFFFFFHGHJJJJJJ
@SRR948304.5196 UNC1
GGAACGCCTACTGGTACTAT
+
CCCCFFFFHHHJGHIJJ
@SRR948304.5200 UNC1
CAGGAAGAAGGAGTAGTCCA
+
@@CFFBDDHHBHIFGHH
@SRR948304.5215 UNC1
CCTTCTCAACGACTACTAC
+
@@@FFFFHGDHFIEHIIJ
@SRR948304.5247 UNC1
CTACAGCTTCCGCAAGAACT
+
BB@FFFFHHHFIJJJJJ
@SRR948304.5250 UNC1
CCTGGTTGAACGCTAGATA
+
CCCCFFFFHHHJGHIJJ

```



Geneid	lrrel_kd_1	lrrel_kd_2	lrrel_kd_3	Mov10_oe_1	Mov10_oe_2	Mov10_oe_3
ENSG000002	0	0	0	0	0	1
ENSG000002	0	0	0	0	0	0
ENSG000003	82	86	65	119	106	52
ENSG000002	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000002	0	2	1	0	3	3
ENSG000002	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000003	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000001	0	0	0	0	0	0
ENSG000002	2	0	0	2	1	0
ENSG000002	0	0	0	0	0	0
ENSG000003	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000003	1	1	0	0	1	0



Now we have a bunch of DE genes

Ensgene	Symbol	baseMean	log2FoldChange	lfcSE	Pvalue	padj
ENSG000000000003	TSPAN6	3556.806146 0	-0.441127573	0.0502311 3	2.718202e-19	1.974760e-17
ENSG000000000005	TNMD	26.5780991	-0.001824407	0.1901698 0	9.839496e- 01	9.916255e- 01
ENSG000000000419	DPM1	1647.820357 2	0.357082308	0.0728420 9	2.710035e- 07	2.804880e- 06
ENSG000000000457	SCYL3	525.1703268	0.201839039	0.0872990 9	1.202293e- 02	3.882876e- 02
ENSG000000000460	FIRRM	1157.706614 5	-0.365719321	0.0676305 5	1.773563e- 08	2.288515e- 07

So, we have a list of thousands of differentially expressed genes...

- Visualizing and filtering to prioritize
 - Which genes are the most significant?
 - What about the expression of my favorite gene?

MA plots, for example

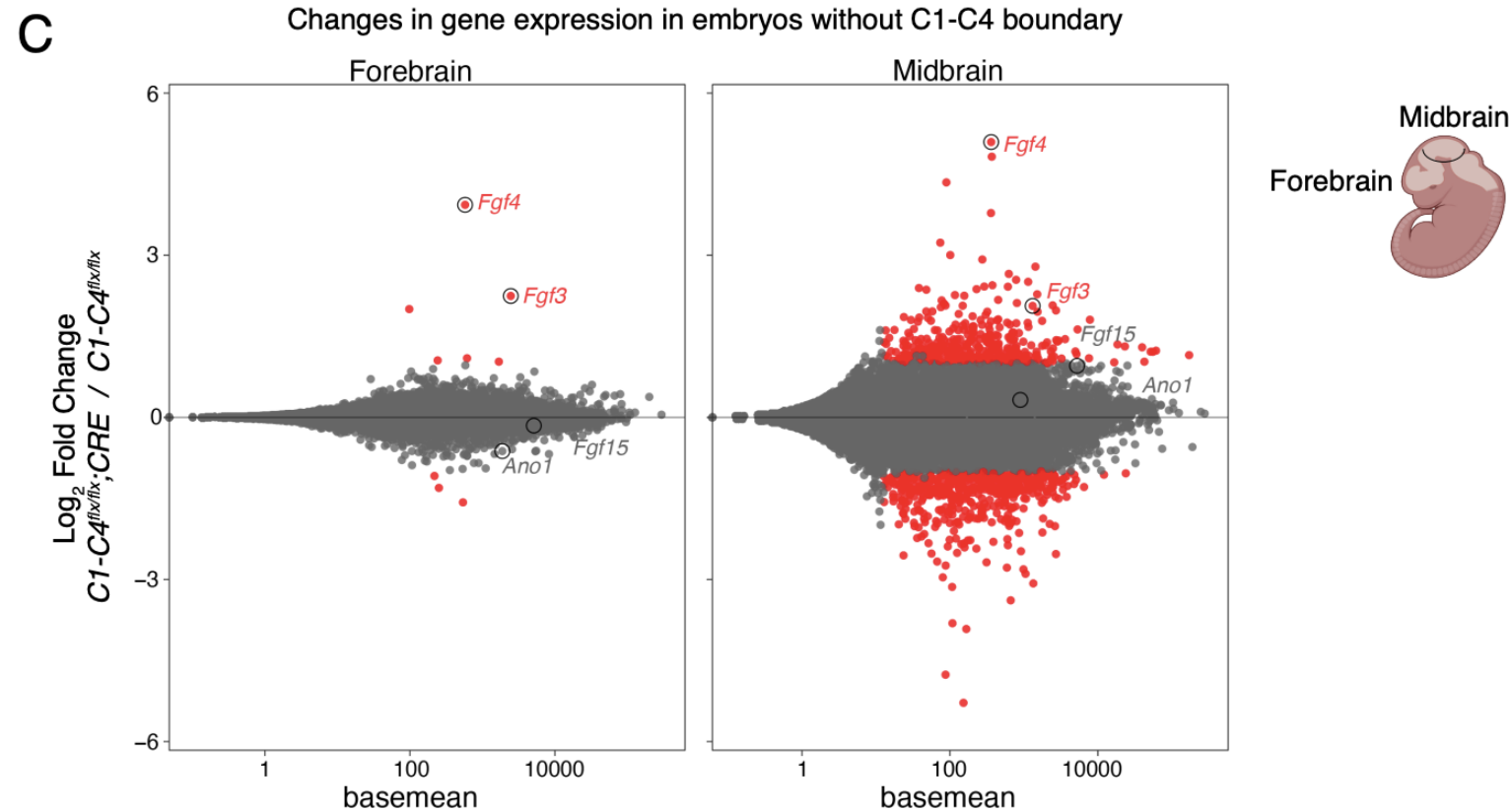
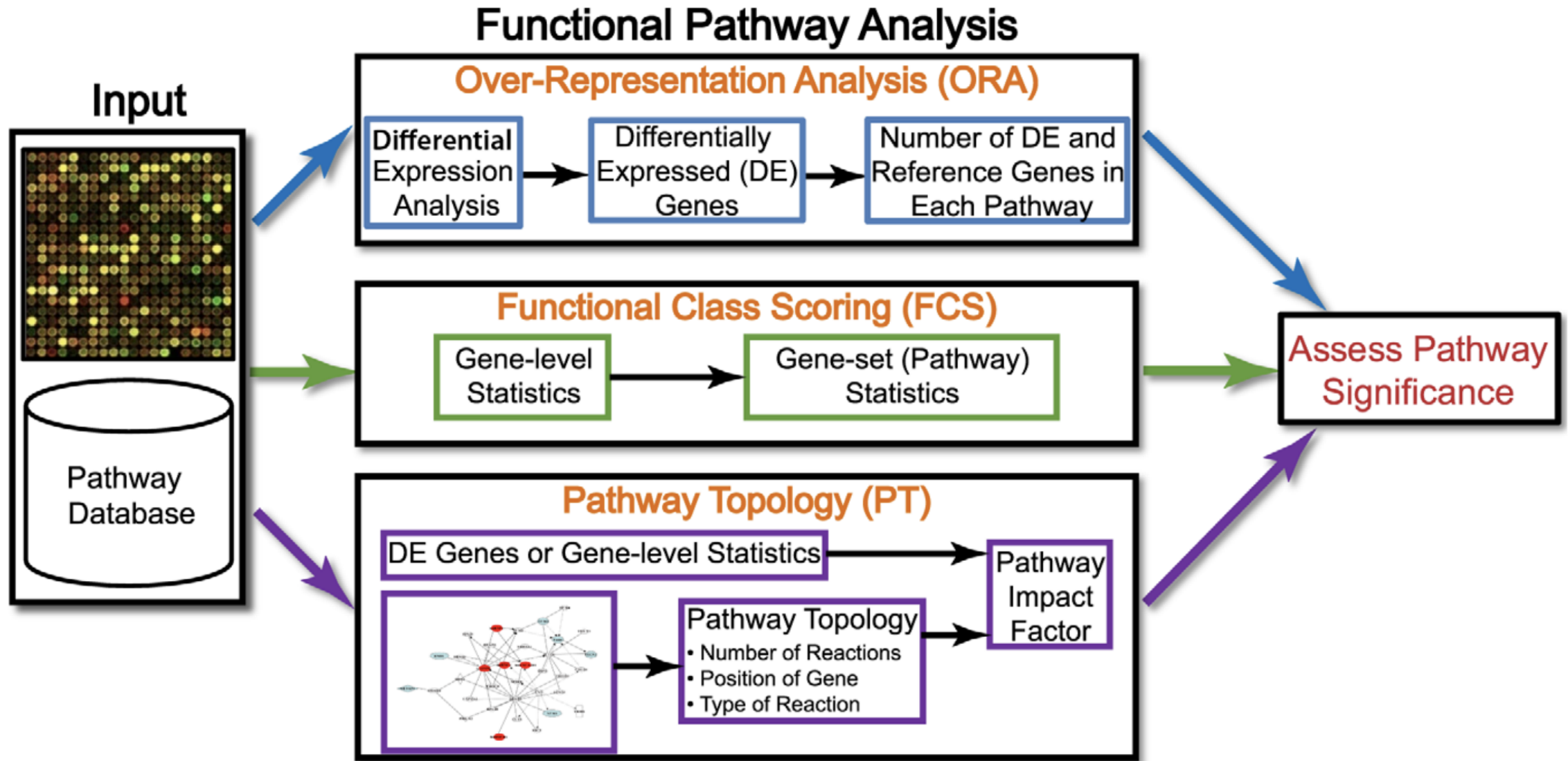


Figure S5 Disruption of the C1-C4 boundary leads to ectopic expression of FGF genes in the brain, related to Figure 4. A Scheme of the breeding and tamoxifen treatment used to generate mice shown in Figure 4. **B** HCR of E9.5 embryos showing higher expression of FGF genes in the midbrain and anterior forebrain in heterozygous embryos with C1-C4 deletion (n=3 for each genotype). **C** MA-plot of E11.5 anterior forebrain and midbrain showing all differentially expressed genes between the two genotypes. Red shows genes that were considered to be differentially expressed (adjusted pvalue<0.1, log2FC>2).

What about the rest of the genes?

- What types of genes are common in my list of overexpressed genes?
 - Annotation with functional information
 - Overrepresentation Analysis (ORA)
 - Gene Set Enrichment Analysis

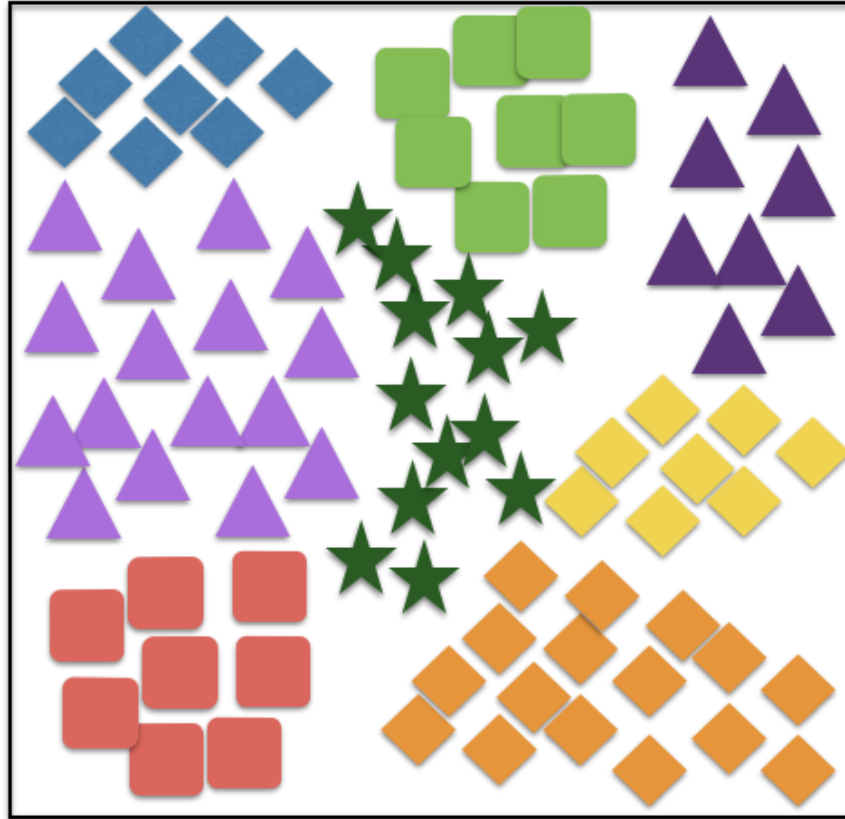


<https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1002375>

Overrepresentation Analysis

Over-representation analysis (ORA) is used to determine which *a priori* defined gene sets are more present (over-represented) in a subset of “interesting” (i.e. DE) genes than what would be expected by chance ([Huang et al., 2009](#)).

All known genes in a species
(categorized into groups)



The “universe”



DEGs

Are red squares overrepresented in the DEGs compared to the full set of genes?

Genes categories	Organism-specific Background	DE results	Over-represented?
Functional category 1	35/13000	25/1000	Likely
Functional category 2	56/13000	4/1000	Unlikely
Functional category 3	90/13000	8/1000	Unlikely
Functional category 4	15/13000	10/1000	Likely
...			
...			

Hypergeometric Test

Describes the probability of 25 genes (k) being associated with "Functional category 1", for all genes in our gene list ($n=1000$), from a population of all of the genes in entire genome ($N=13,000$) which contains 35 genes (K) associated with "Functional category 1" -> **get a P-value**

$$P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

Functional category 1	35/13000	25/1000	Likely
-----------------------	----------	---------	--------

<https://systems.crump.ucla.edu/hypergeometric/index.php>

GO Terms

•**Biological process:** refers to the biological role involving the gene or gene product, and could include “transcription”, “apoptosis”.

•**Molecular function:** represents the biochemical activity of the gene product, such activities could include “ligand”, “GTPase”, and “transporter”.

•**Cellular component:** refers to the location in the cell of the gene product. Cellular components could include “nucleus”, “lysosome” etc.

Example:

https://amigo.geneontology.org/amigo/gene_product/UniProtKB:P31314



The network of biological classes describing the current best representation of the “universe” of biology: the molecular functions, cellular locations, and processes gene products may carry out.

[i GO Ontology Overview](#)

[Browse in AmiGO](#)

[Download](#)



Statements, based on specific, traceable scientific evidence, asserting that a specific gene product is a real exemplar of a particular GO class.

[i GO Annotations Overview](#)

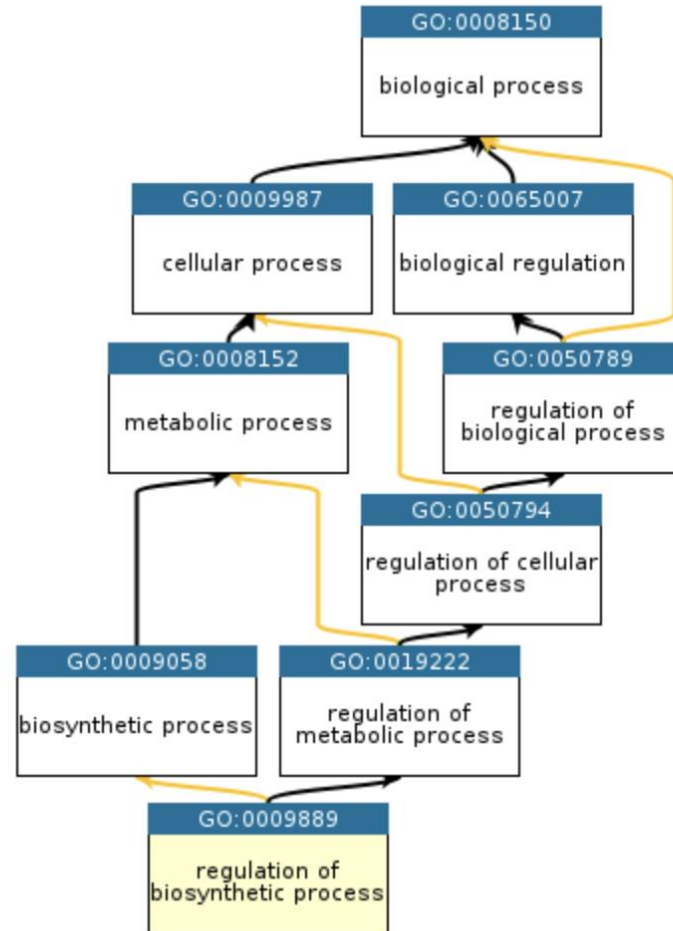
[PAN-GO Functionome](#)

[Browse in AmiGO](#)

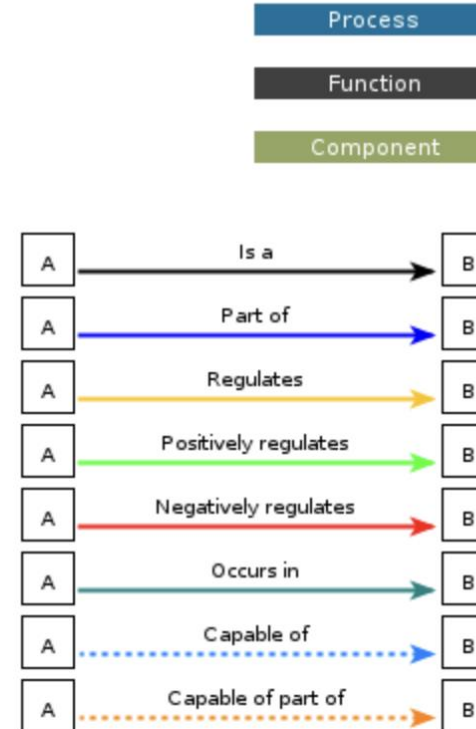
[Download](#)

<https://geneontology.org/docs/guide-go-evidence-codes/>

GO Terms are Hierarchical



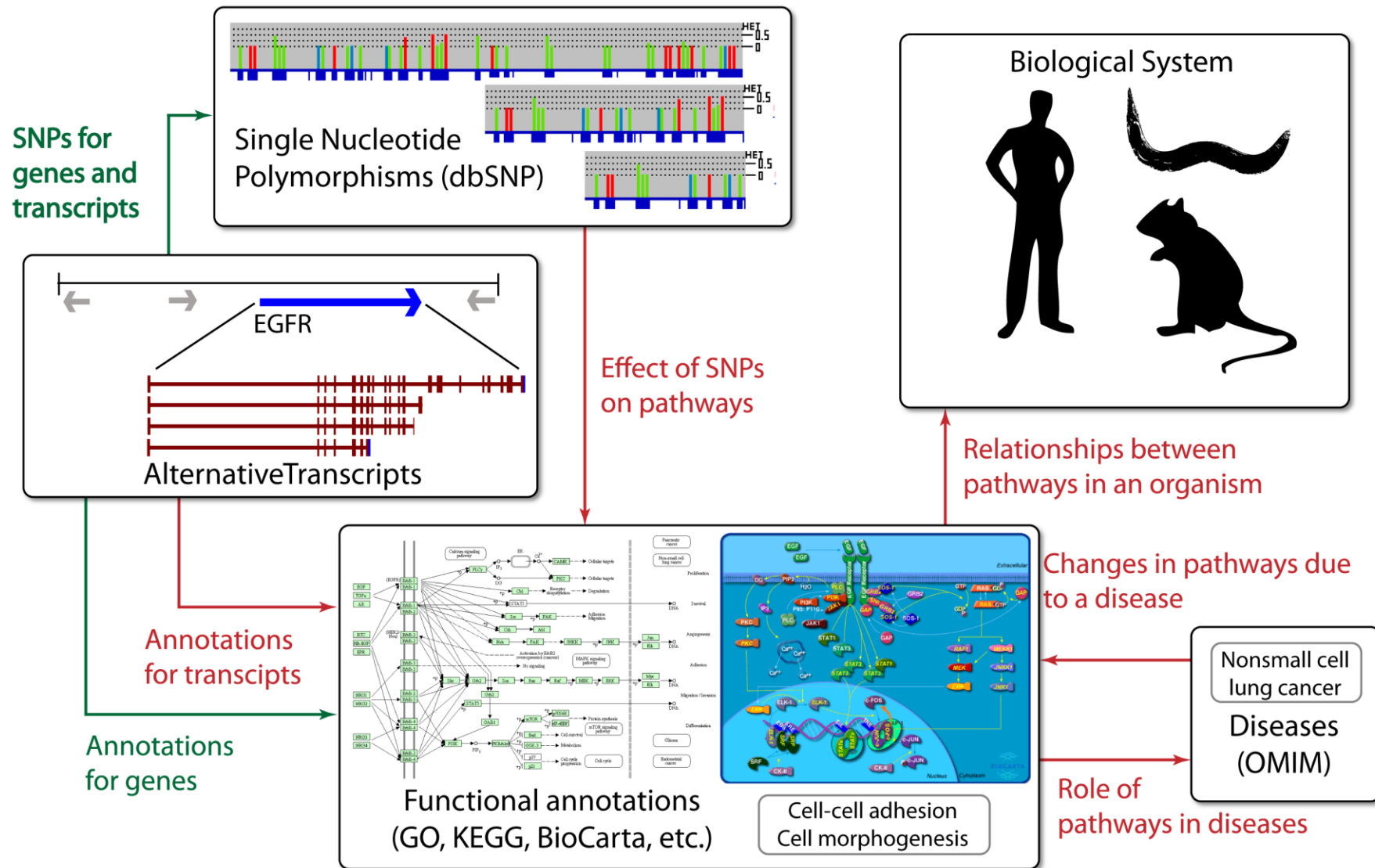
QuickGO - <https://www.ebi.ac.uk/QuickGO>



https://amigo.geneontology.org/amigo/dd_browser

Caveats to ORA and other approaches

- Ignores magnitude of differential expression within the group of DE genes
- Independence of genes from one another? Pathways from one another?
- Quality and availability of annotations



Week 9: Day 1 – Gene Set Enrichment

The **hypothesis behind functional class scoring (FCS) methods**:

- Large changes in individual genes can have significant effects on pathways (and will be detected via ORA methods)
- **Weaker but coordinated changes in sets of functionally related genes (i.e., pathways) can also have significant effects.**
- Thus, rather than setting an arbitrary threshold to identify 'significant genes', **all genes are considered** in the analysis.

Gene set enrichment analysis utilizes the gene-level statistics (log2 fold changes) for all genes to look to see whether gene sets for particular biological pathways are enriched among the large positive or negative fold changes.

SIGNIFICANCE

UPREGULATED

DOWNREGULATED

Ranked list

	Gene symbol	Ranking
1	<i>JAK9</i>	12.9
2	<i>RDH16</i>	11.4
3	<i>ALDOB</i>	11.3
5	<i>AKR1C4</i>	10.7
6	<i>ABCB11</i>	8.6
7	<i>BAAT</i>	3.6
8	<i>SLC2A2</i>	0.0099
9	<i>FCER1G</i>	0.00673
	...	
6532	<i>MyD88</i>	-9
6533	<i>TRIF</i>	-9.37
6534	<i>IRAK1</i>	-15.6

SIGNIFICANT + UPREGULATED

NON-SIGNIFICANT

SIGNIFICANT + DOWNREGULATED

	Gene symbol	Ranking
1	<i>PTPN22</i>	12.9
2	<i>RDH16</i>	11.4
3	<i>IL1B</i>	11.3
5	<i>TNF</i>	10.7
6	<i>ABCB11</i>	8.6
7	<i>SPON2</i>	3.6
8	<i>SLC2A2</i>	0.0099
9	<i>TRIM55</i>	0.00673
	...	
6532	<i>OSM</i>	-9
6533	<i>TLR6</i>	-9.37
6534	<i>ZAP70</i>	-15.6

RANKED LIST



(181 GENES)

interleukin-6 production (GO:0032635)*	PTPN22, IL1B, CSK, TNF, TRAF6, TLR2, RABGEF1, PYCARD, SIRPA, SPON2, TLR7, TLR6, TRIM55, ZG16, TYROBP, UCN, XBP1...
---	--

GENE SET

How does GSEA work?

Step 1: Calculation of enrichment score:

- An enrichment score for a particular gene set is calculated by walking down the list of log2 fold changes and increasing the running-sum statistic every time a gene in the gene set is encountered and decreasing it when genes are not part of the gene set. The size of the increase/decrease is determined by magnitude of the log2 fold change. Larger (positive or negative) log2 fold changes will result in larger increases or decreases. The final enrichment score is where the running-sum statistic is the largest deviation from zero.

Step 2: Estimation of significance:

- The significance of the enrichment score is determined using permutation testing, which performs rearrangements of the data points to determine the likelihood of generating an enrichment score as large as the enrichment score calculated from the observed data

Step 3: Adjust for multiple test correction

After all gene sets are tested, the enrichment scores are normalized for the size of the gene set, then the p-values are corrected for multiple testing.

- The GSEA output will yield the core genes in the gene sets that most highly contribute to the enrichment score. The genes output are generally the genes at or before the running sum reaches its maximum value (eg. the most influential genes driving the differences between conditions for that gene set).

	Gene symbol	Ranking
1	<i>PTPN22</i>	12.9
2	<i>RDH16</i>	11.4
3	<i>IL1B</i>	11.3
5	<i>TNF</i>	10.7
6	<i>ABCB11</i>	8.6
7	<i>SPON2</i>	3.6
8	<i>SLC2A2</i>	0.0099
9	<i>TRIM55</i>	0.00673
	...	
6532	<i>OSM</i>	-9
6533	<i>TLR6</i>	-9.37
6534	<i>ZAP70</i>	-15.6

RANKED LIST

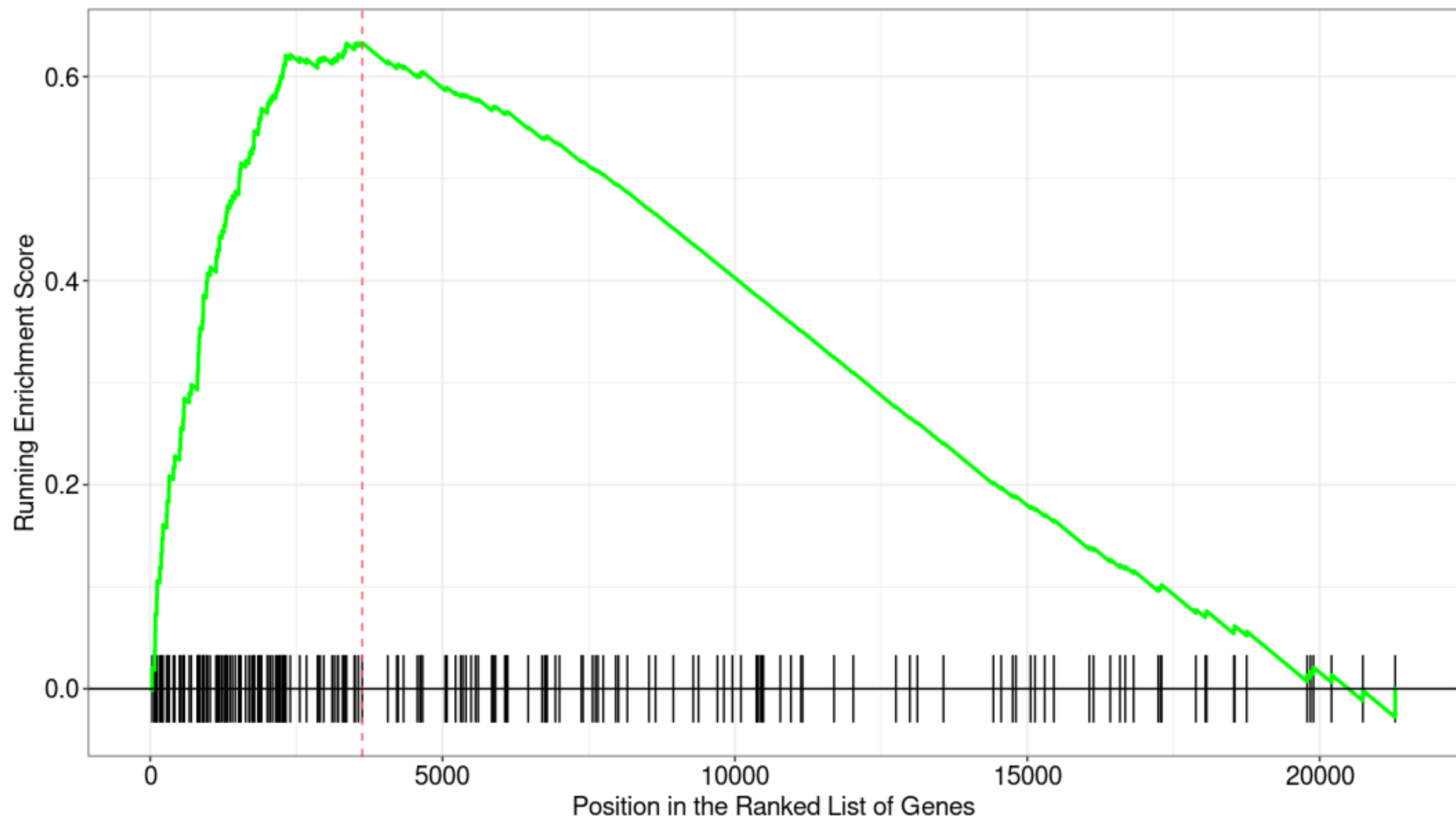


(181 GENES)

interleukin-6 production (GO:0032635)*	PTPN22, IL1B, CSK, TNF, TRAF6, TLR2, RABGEF1, PYCARD, SIRPA, SPON2, TLR7, TLR6, TRIM55, ZG16, TYROBP, UCN, XBP1...
---	--

GENE SET

HALLMARK_HYPOXIA



How does GSEA work?

Step 1: Calculation of enrichment score:

- An enrichment score for a particular gene set is calculated by walking down the list of log2 fold changes and increasing the running-sum statistic every time a gene in the gene set is encountered and decreasing it when genes are not part of the gene set. The size of the increase/decrease is determined by magnitude of the log2 fold change. Larger (positive or negative) log2 fold changes will result in larger increases or decreases. The final enrichment score is where the running-sum statistic is the largest deviation from zero.

Step 2: Estimation of significance:

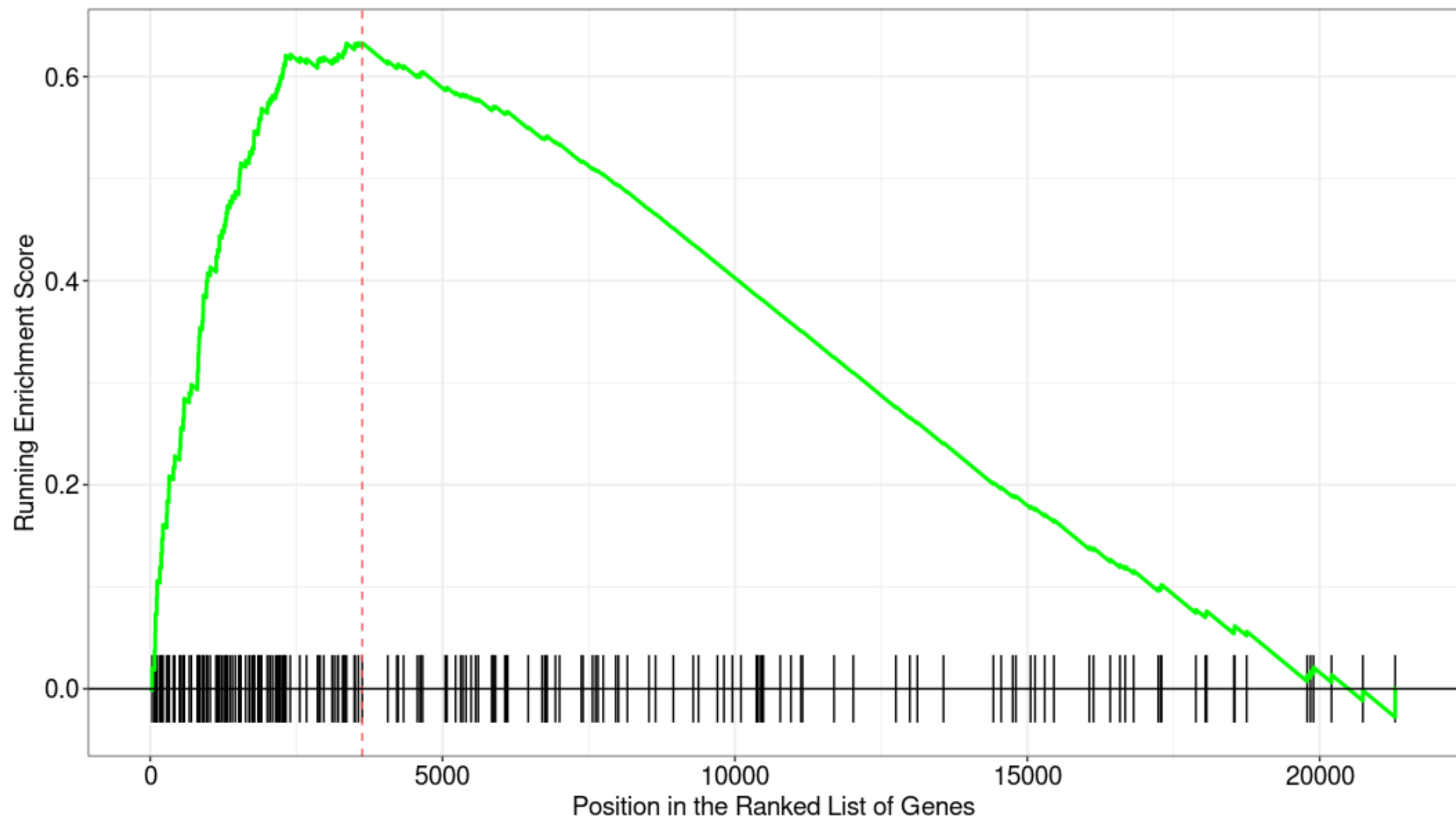
- The significance of the enrichment score is determined using permutation testing, which performs rearrangements of the data points to determine the likelihood of generating an enrichment score as large as the enrichment score calculated from the observed data

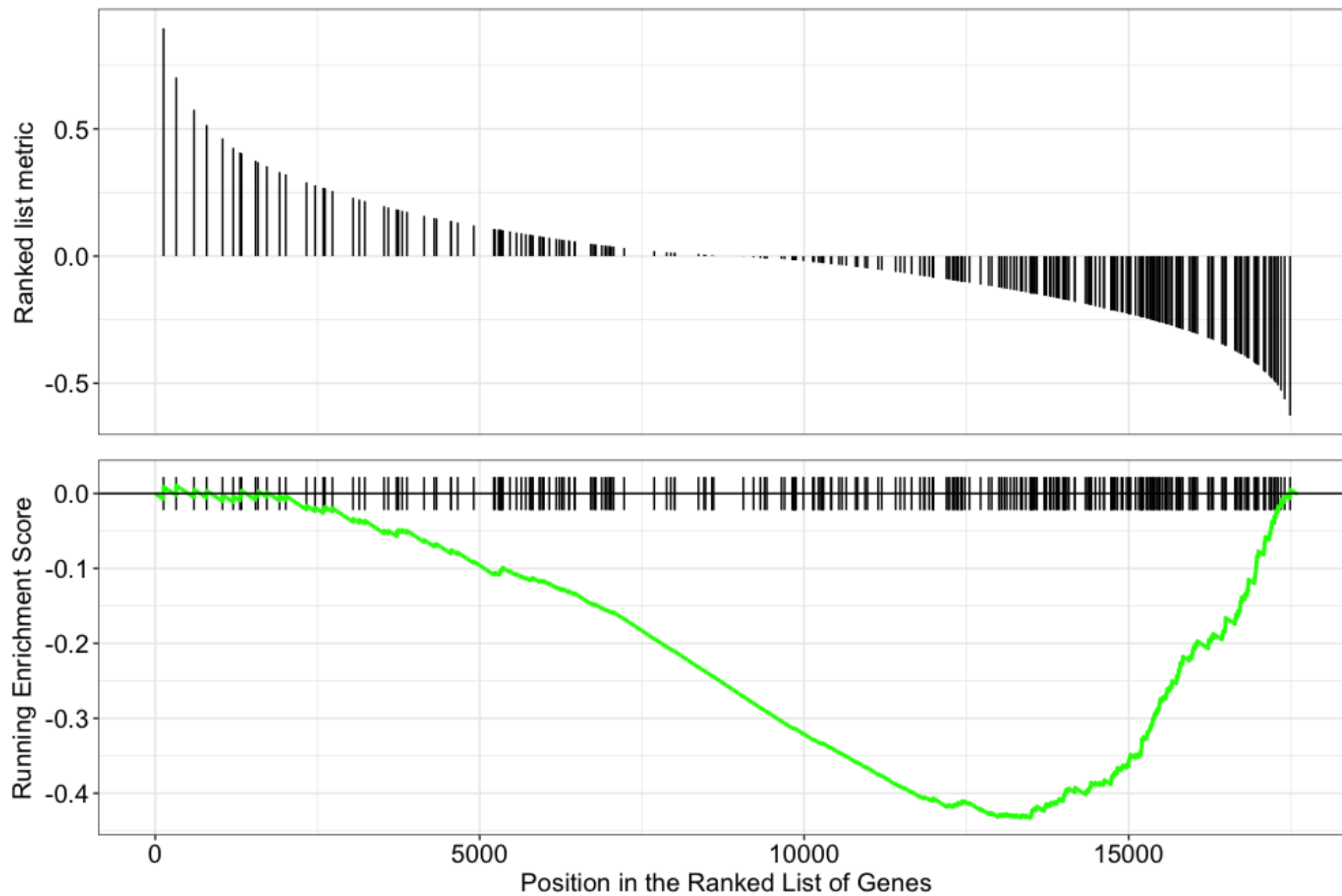
Step 3: Adjust for multiple test correction

After all gene sets are tested, the enrichment scores are normalized for the size of the gene set, then the p-values are corrected for multiple testing.

- The GSEA output will yield the core genes in the gene sets that most highly contribute to the enrichment score. The genes output are generally the genes at or before the running sum reaches its maximum value (eg. the most influential genes driving the differences between conditions for that gene set).

HALLMARK_HYPOXIA





	Gene symbol	Ranking
1	<i>PTPN22</i>	12.9
2	<i>RDH16</i>	11.4
3	<i>IL1B</i>	11.3
5	<i>TNF</i>	10.7
6	<i>ABCB11</i>	8.6
7	<i>SPON2</i>	3.6
8	<i>SLC2A2</i>	0.0099
9	<i>TRIM55</i>	0.00673
	...	
6532	<i>OSM</i>	-9
6533	<i>TLR6</i>	-9.37
6534	<i>ZAP70</i>	-15.6

RANKED LIST



(181 GENES)

interleukin-6 production (GO:0032635)*	PTPN22, IL1B, CSK, TNF, TRAF6, TLR2, RABGEF1, PYCARD, SIRPA, SPON2, TLR7, TLR6, TRIM55, ZG16, TYROBP, UCN, XBP1...
---	--

GENE SET

[illegible]

RANKED LIST



(629 GENES)

Cellular division (GO:0051301)*	ACTR2, ANAPC4, BECN1, CAT, CALM3, ETV5, EXOC8, ETV3, FAM83D, JTB, KAT14, KIF14, MEN1, MEN2, NANOG, MYC, P53...
------------------------------------	--

GENE SET

DOWNREGULATED PATHWAYS WILL LOOK LIKE THIS...

[illegible]

RANKED LIST

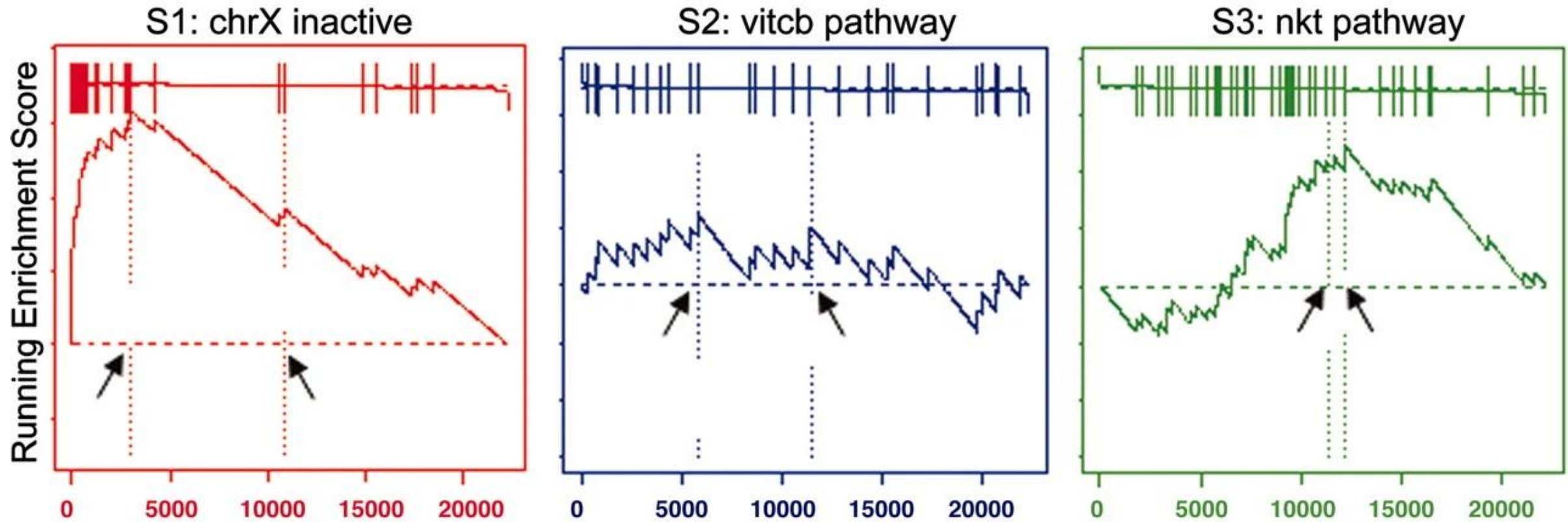


(629 GENES)

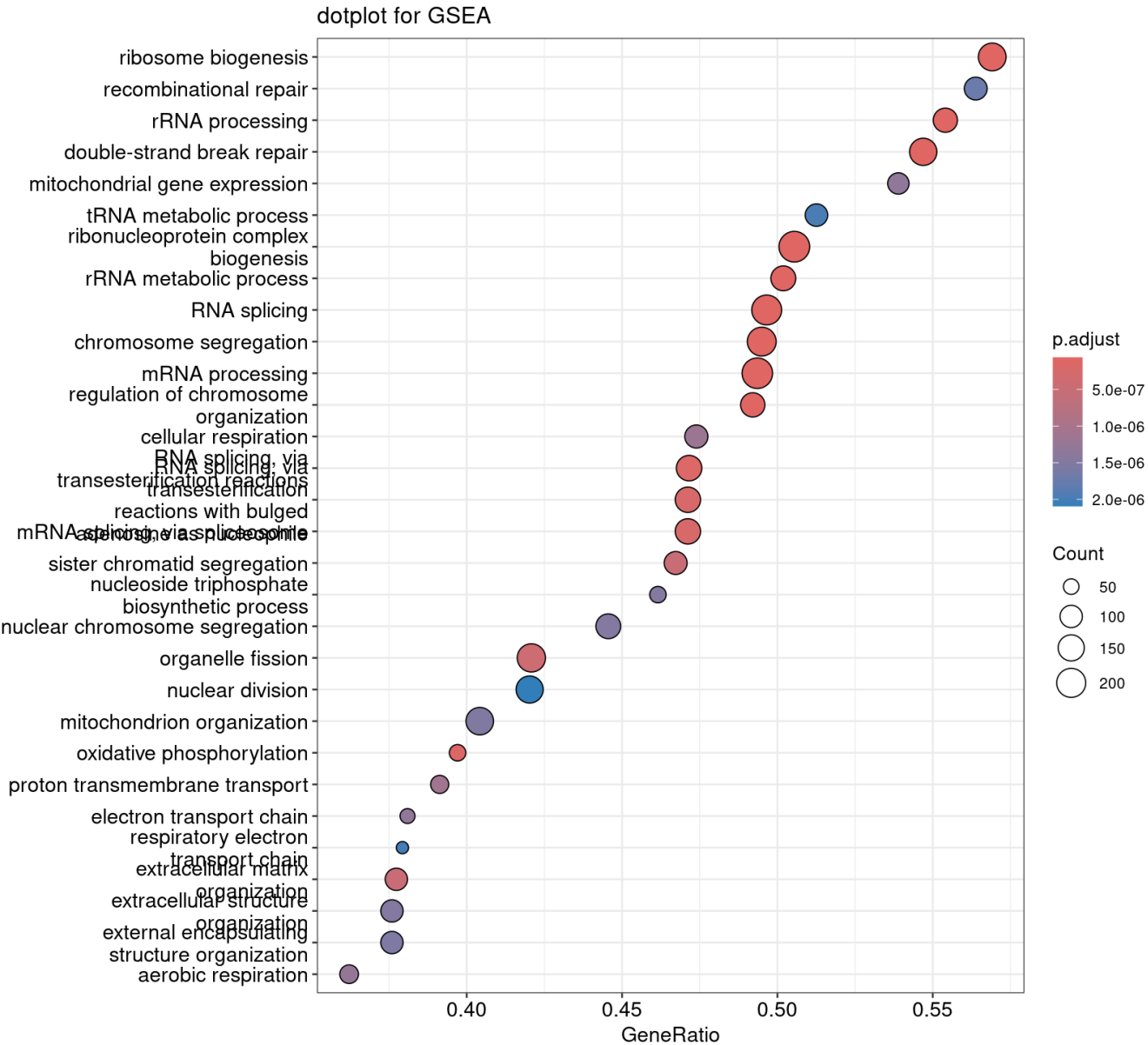
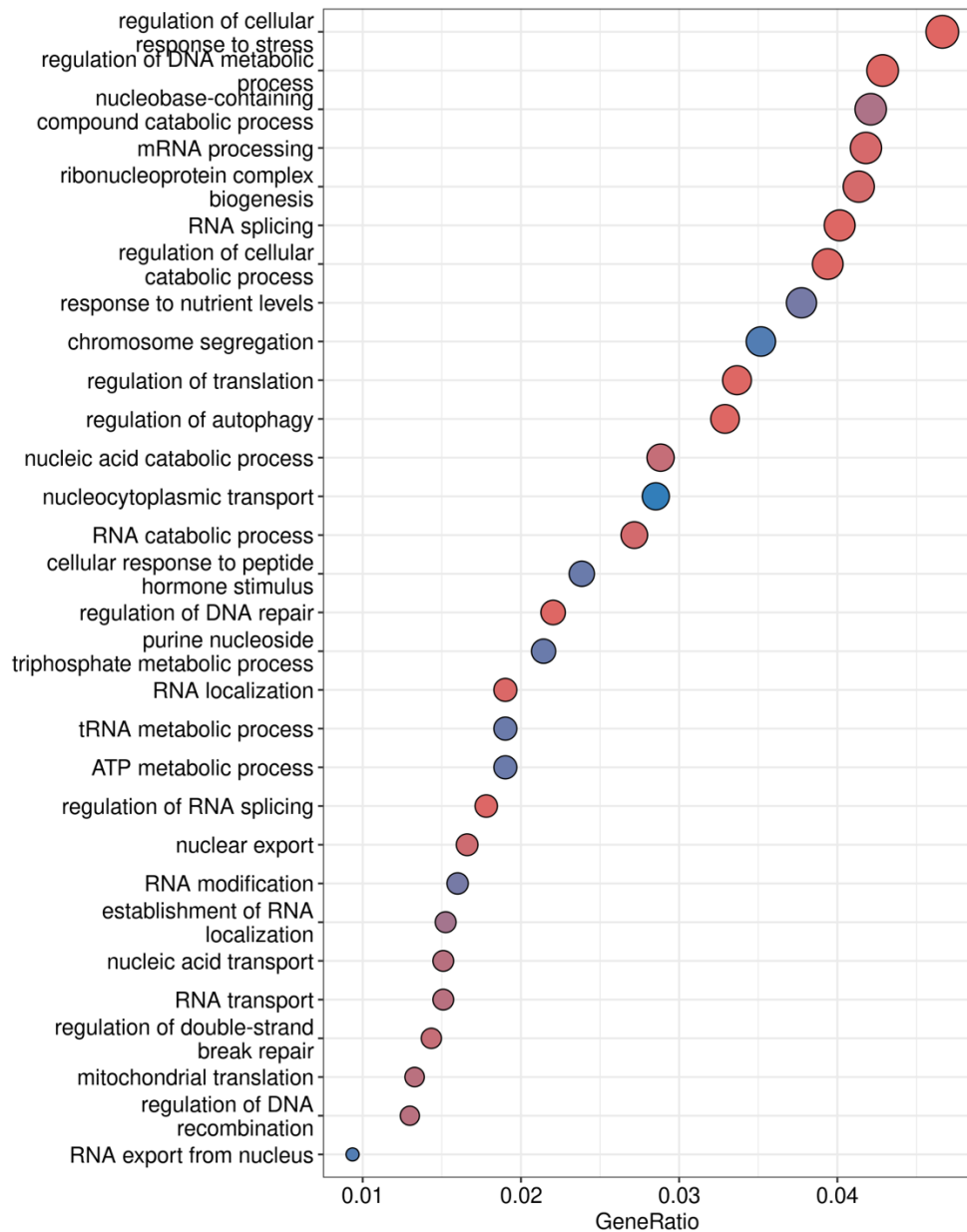
Heart contraction (GO:0051301)*	ACTR2, ANAPC4, BECN1, CAT, CALM3, ETV5, EXOC8, ETV3, FAM83D, JTB, KAT14, KIF14, MEN1, MEN2, NANOG, MYC, P53...
------------------------------------	--

GENE SET

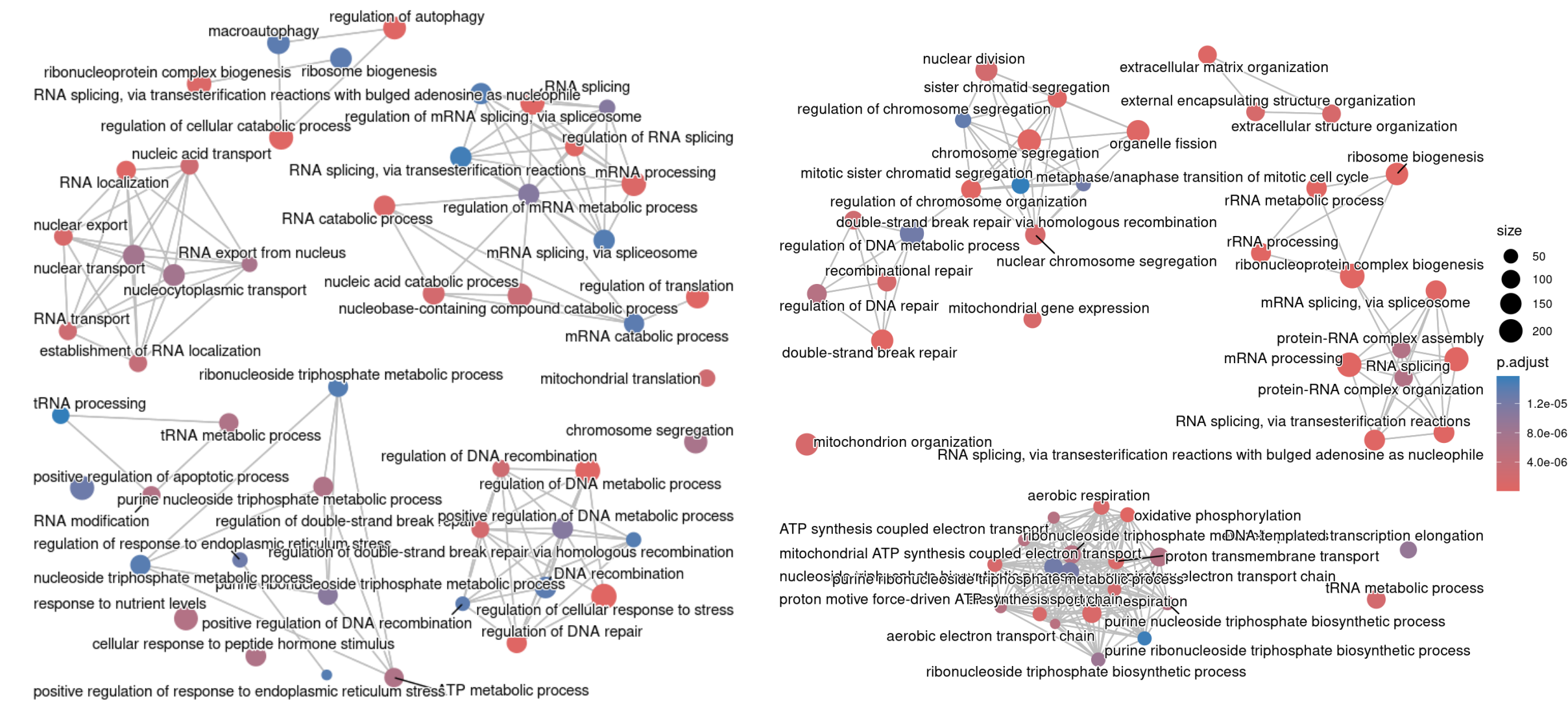
PATHWAYS THAT ARE NOT OVERREPRESENTED WILL LOOK LIKE THIS...



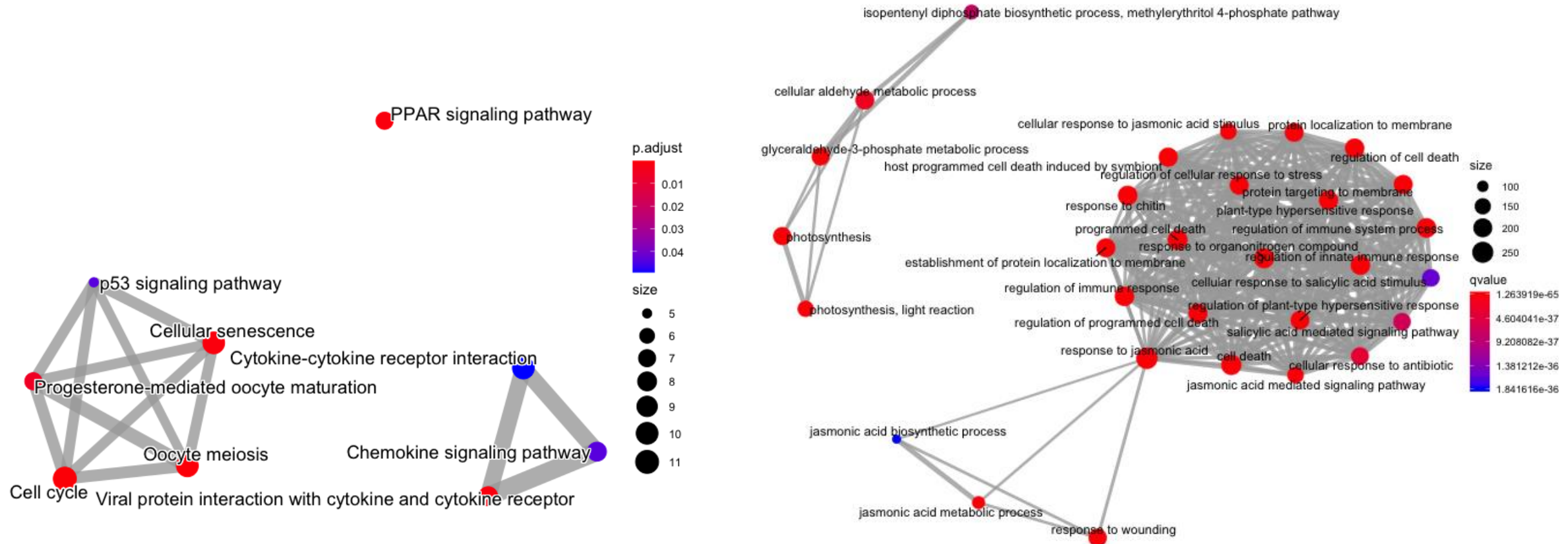
r. The distribution of three gene sets, from the C2 functional collection, in the list of genes in the male/female lymphoblastoid cell line example ranked by their correlation with gender: S1, a set of chromosome X inactivation genes; S2, a pathway describing vitamin c import into neurons; S3, related to chemokine receptors expressed by T helper cells. Shown are plots of the running sum for the three gene sets: S1 is significantly enriched in females as expected, S2 is randomly distributed and scores poorly, and S3 is not enriched at the top of the list but is nonrandom, so it scores well. Arrows show the location of the maximum enrichment score and the point where the correlation (signal-to-noise ratio) crosses zero.



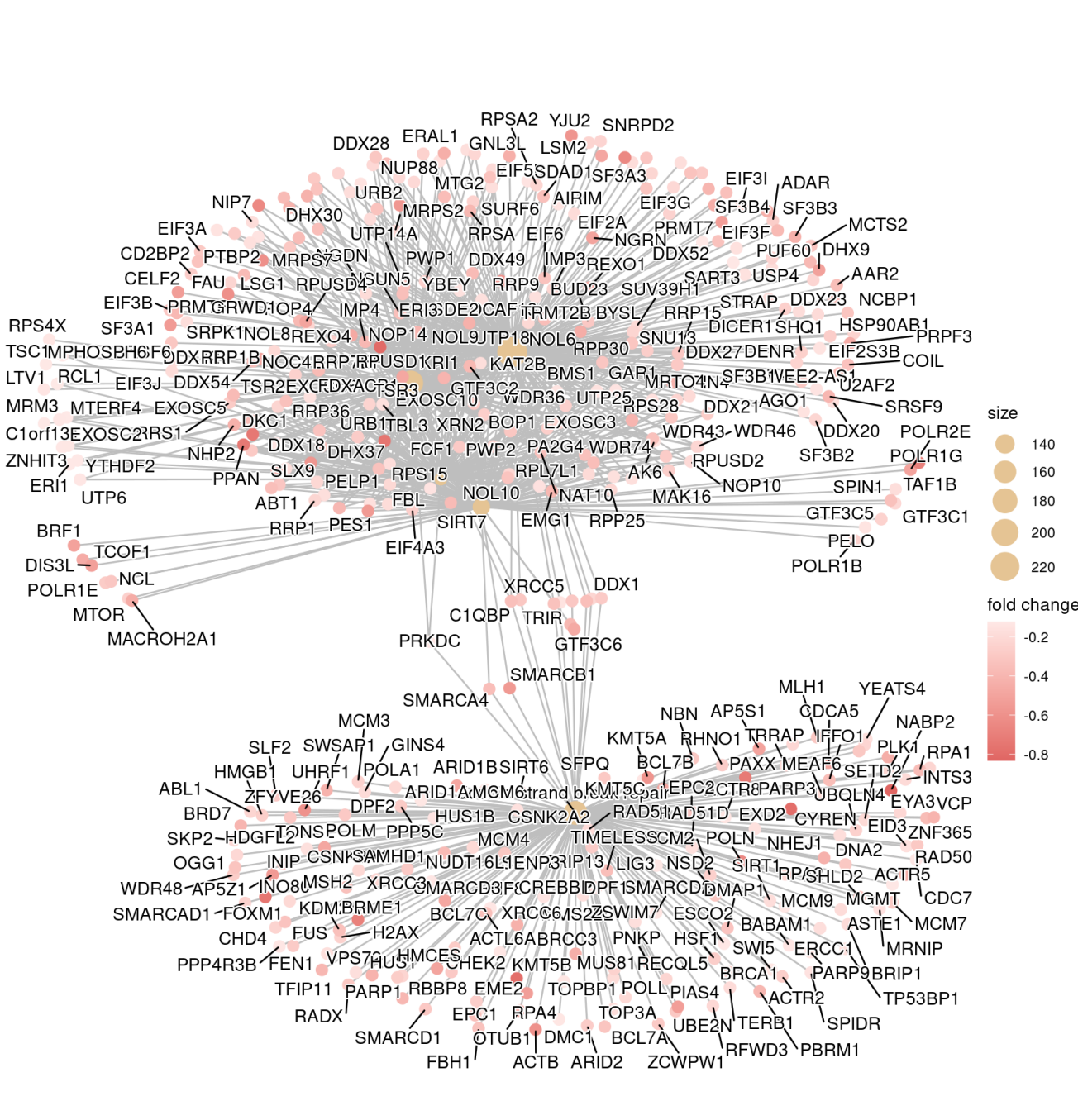
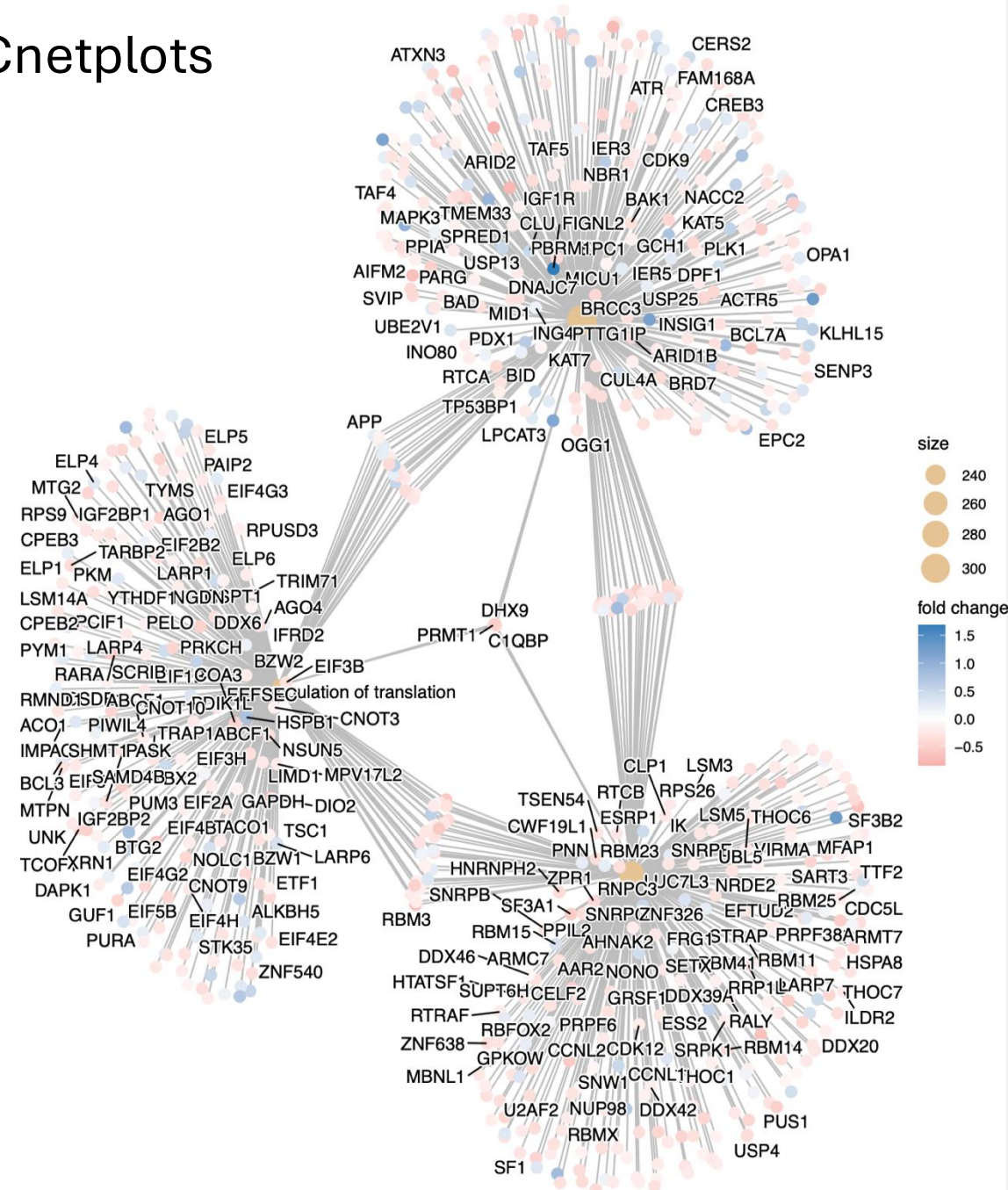
Emapplots



Really depends on the size of your data and plotting parameters



Cnetplots



More useful cnetplots

